

REFERENCE  *A. Bhardwaj & K. Kaushik (2023) – Practical Digital Forensics*

- ✓ Definitions
- ✓ Goals
- ✓ Cybercrime and sources
- ✓ Digital evidence (types and locations)
- ✓ Forensics categories
- ✓ Data analysis
- ✓ Users
- ✓ Investigation types
- ✓ Forensics readiness
- ✓ Chain of custody
- ✓ Examination process

1. Introduction

Digital forensics is an Important part of modern cyber investigations. As technology becomes central to daily life and business, **cybercrime** has increased rapidly. To investigate these crimes, we need **scientific methods** to collect and analyze **digital evidence**.

Example:

Every second, emails, transactions, and logs are created. When a crime happens (like data theft or fraud), this information becomes vital evidence.

2. Defining Digital Forensics

Definition:

A branch of forensic science that applies **scientific techniques** to **identify, collect, preserve, analyze, and present** digital evidence in a court of law.

Purpose:

To find out:

- **What happened**
- **When it happened**
- **Who was responsible**

Scope:

- Computers (PCs, laptops)
- Mobile devices
- Internet of Things (IoT)
- Storage media (USBs, SSDs, CDs)

Principles:

- Evidence must remain **authentic**
- Process must be **repeatable**_ A repeatable process is one that can be done again by another investigator and will give the same findings.
- Results must be **verifiable** and **legally admissible** •

3. Goals of Digital Forensics

1. Preserve Evidence

- Prevent alteration or damage

2. Follow Legal Procedures

- Ensure evidence is acceptable in court

3. Assign Responsibility

- Identify the actor behind an activity

4. Understand Incidents

- Detect data breaches and impacts o

5. Document Clearly

- Create formal, understandable reports

6. Support Legal Process

- Present evidence and testify in court

Key Point:

Integrity and accuracy are more important than speed.

4. Defining Cybercrime



Definition:

Any **illegal activity** carried out **against or using** a computer or network.



Main Motives:

- Financial gain
- Spy
- Disruption
- Revenge



Common Examples:

- Phishing
- Identity theft
- Ransomware •
- Unauthorized access
- Denial of Service (DoS) • •
- Data theft and sabotage

5. Sources of Cybercrime

◆ **1. Insider Threats**

- Employees, ex-staff, or contractors
- Already have authorized access
- Often harder to detect

◆ 2. External Attackers

- Hackers or cybercriminals
- Use phishing, malware, or stolen credentials
- May cooperate with insiders •

🧠 Example:

A disgruntled employee leaks data, or an outside hacker breaches the system using stolen passwords.

6. Computers in Cybercrimes

✖ Three Roles:

1. Computer as a Weapon

- Used to attack others (e.g., DoS, ransomware)

2. Computer as a Target

- Victim of an attack (e.g., hacking or theft)

3. Computer as a Tool

- Used to plan or assist crime (e.g., communication, storage)

🧠 Example:

A hacker uses one system to launch attacks (weapon), steals data from another (target), and hides stolen files on a third (tool).

7. Categories of Digital Forensics

💻 1. Computer Forensics

- Analysis of PCs, laptops, and drives
- Recover deleted files and logs

2. Mobile Forensics

- Extract data from smartphones and SIMs
- Recover messages, call logs, GPS data

3. Network Forensics

- Capture and analyze network traffic

The instruction “**Capture and analyze network traffic**” means you must **collect (capture)** data moving across a network and then **examine (analyze)** it to find useful information — like connections, websites visited, files transferred, or suspicious activity.

- Detect intrusions or data leaks •

• 4. Database Forensics

- Investigate SQL logs and transactions
- Detect data manipulation or fraud

5. Other Subfields

- **Cloud Forensics:** Virtual servers, SaaS
- **IoT Forensics:** Smart sensors and devices
- **Memory Forensics:** RAM and live data

8. Forensic Data Analysis

Focus:

Structured corporate data like:

- Databases
- Financial records
- Audit logs •
-  **Used For:**
- Fraud detection •

- Policy violation tracking •
- Pattern recognition

⚙️ **Techniques:**

- Log correlation=relationship •

The term “**Log correlation**” means **combining and analyzing logs from different sources** to find **patterns, relationships, or evidence** that a single log alone cannot show.

- Timeline reconstruction •

The phrase “**Timeline reconstruction**” means **building a chronological sequence of digital events** (what happened, when, and how) during an investigation.

- Behavior analysis

The phrase “**Behavior analysis**” in **digital forensics** means studying **how a user, system, or attacker behaves** by examining **digital evidence** — to understand **patterns, intentions, and anomalies**.

9. Digital Forensic Users



Law Enforcement:

Investigate criminal cases and present evidence in court.



Corporate / Civil Litigation:

Investigate fraud, theft, or employee misconduct.



Intelligence Agencies:

National security, terrorism, and espionage.



Legal Professionals:

Use digital evidence in lawsuits and trials.



Key Idea:

Digital forensics supports both **criminal** and **civil** justice systems.

10. Investigation Types

◆ **Public Investigations**

- Run by **law enforcement**
- Based on **legal frameworks**
- Example: Police cybercrime unit investigating hacking

◆ **Private Investigations**

- Done by **organizations**
- For internal issues: data leaks, policy violations •
- May later support court cases

11. Forensics Readiness

Definition:

The capability to collect, preserve, and use digital evidence **quickly and legally** when incidents occur.

Goals:

- Reduce investigation cost and time
- Ensure compliance with law
- Support faster response
- Detect threats early •

• **Measures:**

- Logging and monitoring
- Data keep policy •
- Incident response plan
- Secure evidence storage

 **Think:** Preparedness = faster and more accurate investigations.

12. Types of Digital Evidence

User-Created Data:

- Documents, images, emails, backups, encrypted files

Machine-Created Data:

- Logs, registry, browser history, metadata, IP/MAC addresses •

Important:

Both human and system data are essential to reconstruct events.

13. Locations of Digital Evidence

Devices:

- PCs, laptops, servers
- Smartphones, IoT devices
- USBs, HDDs, SSDs
- Cloud storage

Network Components:

- Routers, switches, modems
- Firewalls, DVRs, GPS devices

Explain:

Evidence is everywhere — investigators must know **where to look**.

14. Chain of Custody

Definition:

A record that **tracks the evidence** from collection to presentation..

Includes:

- Who collected and handled the evidence
- When and how it was stored
- Tools used for analysis

Goal:

To keep integrity and acceptability in court.

Evidence can be **rejected** in legal proceedings

15. Examination Process

Phases:

1. Search & Seizure

- Identify and collect devices legally
- Capture volatile memory if possible

2. Acquisition

- Create **forensic image** (bit-by-bit copy)
- Use **write-blockers** to avoid tampering

3. Analysis

- Use forensic tools (EnCase, FTK, Autopsy)
- Recover deleted data, logs, hidden files

4. Reporting

- Write clear, non-technical report
- Submit evidence copies to court

Explain:

Each phase must be documented to ensure credibility.

Lecture: Chapter 2 – Essential Technical Concepts

Practical Digital Forensics (2023)

Authors: A. Bhardwaj & K. Kaushik

Purpose: To provide a foundational understanding of computing concepts essential for digital forensics investigations.

1. Introduction

A digital forensic investigation requires a strong understanding of fundamental computing concepts. Forensic examiners must know how computers store, process, and represent data, as well as how digital files are structured and how storage media work. This knowledge ensures that digital evidence is acquired, preserved, and analyzed accurately without data loss or tampering.

This chapter explores key technical areas:

- Number systems and encoding
- File structure and metadata
- Hashing for evidence verification
- System memory and storage
- File systems and slack space
- Cloud computing environments
- IP addressing basics

These topics form the technical foundation necessary for forensic professionals to locate, recover, and analyze digital evidence in a forensically sound manner.

2. Number Systems in Computing

Computers represent and manipulate data using different number systems. Understanding these systems is essential for interpreting raw data, file headers, and memory addresses.

2.1 Decimal (Base-10)

- The decimal system is the one we use in daily life.
- It contains ten digits: 0–9.
- Each position represents a power of 10.

- Example: $4,567 = (4 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (7 \times 10^0)$

2.2 Binary (Base-2)

- Computers operate in binary, using only **0** and **1**.
- Each binary digit is called a **bit**; 8 bits = 1 byte.
- Binary values represent electrical signals: 0 (off) and 1 (on).

- Example: $1101 = (1 \times 8) + (1 \times 4) + (0 \times 2) + (1 \times 1) = 13$

Understanding binary helps forensic analysts interpret raw data and memory dumps.

2.3 Hexadecimal (Base-16)

- Used to simplify binary representation.
- Contains 16 symbols: 0–9 and A–F.
- Commonly used in memory addresses, file signatures, and hash values.

- Example: $0x1A = (1 \times 16) + (10) = 26$

Hexadecimal is a more compact مضغوط form of binary and frequently appears in forensic analysis tools and raw data.

2.4 Base64 Encoding

- Represents binary data as ASCII text.
- Used for transmitting binary files (e.g., images, email attachments) over text-based protocols.
- Example: The phrase “Many hands make light work” encoded in Base64 becomes TWFueSBoYW5kcyBtYWtlIGxpZ2h0IHdvcmsu.

Forensic analysts must recognize Base64-encoded data to decode hidden or transferred content.

3. Character Encoding Schemas

Computers use encoding schemes to convert binary data into readable text.

3.1 ASCII (American Standard Code for Information Interchange)

- Uses 7 bits to represent 128 characters (letters, digits, punctuation).
- Example: Letter **A** = 65 in decimal = 01000001 in binary.
- Limitation: Cannot represent all international characters.

3.2 Unicode

- Global standard for encoding text across all languages.
- Common formats:
 - **UTF-8**: Variable-length encoding; backward-compatible with ASCII.
 - **UTF-16** and **UTF-32**: Fixed-length encodings for broader symbol sets.
- Essential in forensic investigations involving multilingual **متعددة اللغات** data.

4. File Carving and File Structures

4.1 File Carving نحت الفايل

- **Definition**: Recovering files based on content rather than file system metadata.
- Used when file tables are missing, corrupted, or deleted.
- Identifies files using **headers** (beginning bytes) and **footers** (ending bytes).
- Example: A JPEG file starts with FFD8 and ends with FFD9.

Forensic use: Recover deleted images or documents from unallocated space.

4.2 File Structure

A typical file consists of:

- **Header**: Identifies file type (magic number).
- **Body**: The actual data.
- **Footer**: Indicates end of file.

Note: File extensions (e.g., .docx) can be changed to mislead investigators **تضليل المحققين**, but headers reveal true type.

Forensic tools like **Hex Editors** (wxHexEditor, PSPad, Free Hex Editor Neo) allow inspection **تفتيش** of file headers.

5. Digital File Metadata

5.1 Definition

Metadata is **data about data** — additional information stored with a file describing its properties.

5.2 Common Metadata Elements

- Author name
- Creation and modification dates
- Device name or ID
- GPS coordinates (for images)
- Software used
- File size and type

5.3 Importance in Forensics

Metadata reveals:

- When and where a file was created or modified.
- Which device or user created it.
- Possible tampering (e.g., altered timestamps).

Tools for metadata analysis:

- **ExifTool**
- **MediaInfo**
- **XnView**
- **GIMP**
- **PDF Metadata Editor**

Forensic analysts often extract metadata to reconstruct event timelines and user actions.

6. Timestamp Analysis

Files contain three major timestamps:

- **Created** – when the file was first generated.
- **Modified** – last change to the content.
- **Accessed** – last time it was opened or viewed.

By comparing timestamps, investigators can build activity timelines. Some suspects may alter timestamps; forensic tools can detect inconsistencies and reveal tampering.

7. Hash Analysis

7.1 Definition

A **hash** is a unique, fixed-length string derived from file content using a cryptographic algorithm. Used to verify **integrity** and **authenticity** of digital evidence.

7.2 Common Algorithms

- **MD5** (Message Digest)(128-bit)
- **SHA-1** (Secure Hash Algorithm)(160-bit)
- **SHA-256** (256-bit)

7.3 Applications in Forensics

- Validate evidence before and after acquisition.
- Identify duplicate or known files using hash databases.
- Detect file tampering or corruption.

7.4 Tools

- **HashMyFiles** (NirSoft)
- **Febooti Hash & CRC**
- **Windows PowerShell:**
 - `Get-FileHash filename -Algorithm SHA256`

Hashes are like **digital fingerprints**; if the file changes, its hash changes.

8. System Memory

8.1 Types

1. **Volatile Memory** – Data lost when power is off
 - Example: **RAM (Random Access Memory)**
2. **Non-Volatile Memory** – Data retained after power-off
 - Examples: **ROM, Hard Drives, Flash Drives**
 -

8.2 RAM

- Temporarily stores data of running programs.
- Contains:
 - Active processes
 - Browser history
 - Passwords
 - Decrypted data
- Crucial ضروري for live forensic analysis (e.g., malware running in memory).

Types of RAM:

- **D(Dynamic)RAM** – requires constant refresh
- **S(Static)RAM** – faster, used in CPU cache

8.3 ROM

- Stores firmware (e.g., BIOS).
- Non-editable during normal operations.
- Types:
 - **PROM** – programmable once
 - **EPROM** – erasable with UV (ultraviolet) light
 - **EEPROM** – erasable electronically

9. Secondary Storage

9.1 Hard Disk Drives (HDD)

- Magnetic storage with spinning platters.
- Data stored in **tracks**, **sectors**, and **clusters**.
- **Slack space**: unused area in clusters that may contain remnants of deleted files.

9.2 Solid State Drives (SSD)

- Flash-based, faster, no moving parts.

- Limited write cycles but improving over time.
- Used in modern laptops and forensic workstations.

9.3 Backup Storage

- External drives, USBs, optical discs, cloud storage.
- May contain older versions or backups of incriminating data.

10. File Systems

Define how data is stored and retrieved on storage media.

File System	Platform	Features
FAT32	Windows, portable devices	Simple, lacks security
NTFS	Windows default	Supports permissions, journaling
EXT4	Linux	Stable, supports large volumes
HFS+/APFS	macOS	Efficient, supports snapshots

Understanding file systems is vital for identifying deleted files, slack space, and hidden data.

11. Slack Space

- **Definition:** The unused portion within a file cluster.
- **Significance:** May contain fragments of previous files.
- **Use in Forensics:** Recover remnants بقايا of deleted data.
- **Tool:** Disk Slack Checker (Karen's Power Tools).

12. Cloud Computing Environments

Forensics must adapt to cloud-based storage and services.

12.1 Service Models

1. **SaaS (Software as a Service)** – e.g., Gmail, Office 365
2. **PaaS (Platform as a Service)** – e.g., Azure App Services

3. IaaS (Infrastructure as a Service) – e.g., AWS EC2

Challenges:

- Jurisdiction issues
- Data ownership
- Remote acquisition

13. Networking Concepts

IP Addressing

- Unique identifier for network devices.
- **IPv4** – 32-bit (e.g., 192.168.1.1)
- **IPv6** – 128-bit, supports more devices.

Forensic significance:

- Identify source/destination of communications
- Track network-based crimes

14. Recommended Tools

Purpose	Tools
File Carving	Scalpel, PhotoRec
Metadata Analysis	ExifTool, MediaInfo
Hashing	HashMyFiles, PowerShell
Memory Acquisition	FTK Imager, Magnet RAM Capture
File System Analysis	Autopsy, Sleuth Kit

Lecture: Hard Disks and File Systems

1. Introduction

In digital forensics, understanding how data is stored, structured, and retrieved from storage devices is fundamental. Hard disks and file systems form the backbone of digital evidence collection and analysis. Investigators must understand how files are organized, how operating systems interact with storage, and how deleted or hidden data can be recovered for forensic purposes.

2. Objectives

By the end of this lecture, students will be able to:

- Explain the structure and working of hard disks.
- Describe various file systems (NTFS, FAT, EXT, etc.).
- Understand logical and physical disk structures.
- Identify sources of forensic evidence within disks and file systems.
- Analyze registry files for evidence.

3. Hard Disk Concepts

3.1 Definition

A Hard Disk Drive (HDD) is a non-volatile storage medium that retains data even when the power is off. It consists of spinning platters coated with magnetic material, read/write heads, and a controller for managing data access.

3.2 Structure

- Platters: Circular disks that store data magnetically.
- Tracks: Concentric circles on the platter.
- Sectors: Smallest physical storage units (usually 512 bytes or 4 KB).
- Clusters: Logical groupings of sectors used by the file system.
- Cylinders: Set of tracks aligned vertically across platters.

3.3 Logical Structure

The logical view is how the operating system perceives the disk:

- Partitions: Divisions of the disk (Primary, Extended, Logical).
- File System: Structure for organizing and managing files within a partition.

4. File Systems Overview

A file system defines how data is stored, retrieved, and managed on storage devices. It organizes data into files and directories and maintains metadata.

4.1 Common File Systems

1. FAT (File Allocation Table)
 - Used in older systems and removable drives.
 - Variants: FAT12, FAT16, FAT32.
 - Simple structure but limited file size (4 GB) and partition size.
2. NTFS (New Technology File System)
 - Default in modern Windows systems.
 - Supports large files, file permissions, encryption, and journaling.
 - Key structures: Master File Table (MFT), File Record Segments, and Metadata.
3. EXT (Extended File System)
 - Common in Linux: EXT2, EXT3, EXT4.
 - Supports journaling (EXT3/EXT4), large file sizes, and permissions.
4. HFS+ / APFS (Apple File Systems)
 - Used in macOS.
 - APFS supports snapshots and cloning.
5. exFAT
 - Used for flash drives and cross-platform storage.
 - Larger file support than FAT32.

5. File System Components in Forensics

5.1 Metadata

Stores attributes such as file name, creation date, modification time, permissions, and size.

5.2 Directory Structure

Hierarchical organization of files and folders. Deleted files may still be referenced here until overwritten.

5.3 Slack Space and Unallocated Space

- Slack space: Unused bytes within a cluster; may contain remnants of previous files.
- Unallocated space: Space not currently assigned to any file; can contain recoverable deleted data.

5.4 Journals

Used in journaling file systems to record pending operations; useful for reconstructing recent activities.

6. Hard Disk Forensics

6.1 Goals

- Recover deleted files.
- Identify hidden or encrypted partitions.
- Extract timestamps and metadata.
- Ensure data integrity (hash verification).

6.2 Process

1. Imaging: Create a bit-by-bit copy of the disk using tools (FTK Imager, EnCase, dd).
2. Analysis: Examine partitions, files, and hidden data using forensic tools.
3. Recovery: Use carving and signature analysis to retrieve deleted files.
4. Validation: Compare hash values before and after analysis.

6.3 Tools

- Autopsy/The Sleuth Kit
- FTK Imager

- EnCase
- X-Ways Forensics

7. Windows Registry Analysis

The Windows Registry is a central database that stores system and application settings. It is a rich source of forensic evidence.

7.1 Registry Hives

- SAM – User account info and passwords.
- SYSTEM – Hardware configuration.
- SOFTWARE – Installed applications.
- SECURITY – Security policies.
- NTUSER.DAT – User-specific preferences.

7.2 Evidence in Registry

- User login times.
- Installed programs.
- USB device history.
- Network information.
- Recent documents and searches.

7.3 Tools for Analysis

- RegRipper
- Registry Explorer
- Autopsy (Registry Plugin)

8. Forensic Challenges

- Data Overwriting: Deleted data may be overwritten by new files.
- Encryption and Compression: May obscure **يُحجب** data visibility.
- Anti-Forensics: Use of wiping tools or hidden partitions.
- SSD Wear-Leveling: Makes recovery unpredictable.

9. Conclusion

Understanding hard disks and file systems is crucial in digital forensics. Investigators rely on this knowledge to recover data, analyze evidence, and present findings in a legally admissible manner. Mastery of partition structures, file allocation mechanisms, and registry analysis allows for accurate reconstruction of user activities and system events.

10. Summary Points

- Hard disks store data magnetically using structured layers.
- File systems (FAT, NTFS, EXT) define data organization.
- Forensic analysis includes imaging, examination, and recovery.
- Windows Registry provides vital artifacts for user activity.
- Tools like Autopsy, EnCase, and RegRipper are essential.

Internet and Web Forensics

1. Internet Forensics

A specialized area of digital forensics that investigates crimes or incidents occurring over the Internet, such as:

- online attacks
- suspicious network traffic
- Domain Name System (DNS) manipulation
- social media or web-based activities

Goal: Reconstruct events and produce legally admissible evidence.

الهدف: إعادة بناء الأحداث وإنتاج أدلة مقبولة قانوناً.

2. Digital Footprints البصمات الرقمية

Traces left by a user or system on the Internet:

- **Active footprints:** what you choose to share. (posts, comments, messages)
- **Passive footprints:** what systems collect automatically.(logs, cookies, IPs, metadata)

A **cookie** is a small piece of data that websites save on your browser to remember your actions, preferences, or login information.

These footprints help investigators identify *who did what, when, and how.*

3. Role of AI in Internet Forensics

AI helps investigators by:

- **Anomaly detection:** find unusual login attempts, traffic, or behavior
- **Media forensics:** detecting deepfakes or modified images

- **Predictive analysis: identifying attack patterns before they happen**
- **Automated data processing: handling huge logs, PCAPs, and cloud data**

Note A **PCAP** (Packet Capture) is a file format used to **store network traffic** captured from a network interface.

Goal: AI improves speed and accuracy but does not replace human judgment.

4. Legal Considerations

Investigators must:

- **Collect evidence legally (court order / warrantمذكرة).**
- **Preserve integrity using forensic methods (hashing, write blockers).**
- **Follow cybercrime and privacy laws.**

5. Ethical Considerations

- **Protect privacy: access only data relevant to the investigation.**
- **Keep objectivity and avoid bias.**
- **Use tools responsibly and document all actions transparently.**

6. Key Challenges in Internet Forensics

- **Fast data deletion and volatility**
- **Heavy encryption tools•**
- **Global jurisdiction and legal fenceالاختصاص العالمي والحواجز القانونية**
- **Complex cloud and IoT systems**
- **Anti-forensic techniques (log wiping, fake metadata)**
- **Huge data volume and variety**
- **Tool limitations and lack of standards for cloud/web evidence**

6.1. Global jurisdiction

This means different countries have different laws, and it's not always clear which country's law applies when something happens on the internet.

6.2. Legal fence

These are problems or limits caused by laws, such as:

- **countries not sharing data,**
- **slow legal processes,**
- **privacy rules that stop access to information,**
- **different countries having conflicting laws.**

7. Key Steps in Internet Forensics

- 1. Identification: determine what systems and data are involved**
- 2. Preservation: secure evidence, ensure no tampering**
- 3. Collection: gather logs, PCAPs(Packet Capture), cloud data**
- 4. Analysis: reconstruct events, identify attackers or actions**
- 5. Presentation: write a clear forensic report with conclusions and evidence**

8. WAF(Web Application Firewall) Technologies for Web Security

Block Attacker: stop dangerous requests before they reach your system.

WAFs: block attacks like SQLi, XSS

- **Vulnerability Scanners: find system weaknesses**
- **Password Cracking Tools: test password strength ethically**
- **Fuzzing Tools: ادوات التشويش: detect unknown vulnerabilities**

- **Testing Approaches: black-box and white-box testing for secure applications**

Network Forensics

1. Introduction to Network Forensics

Network forensics is a branch of digital forensics that focuses on monitoring, capturing, and analyzing network traffic for the purpose of investigations, security monitoring, and legal evidence. Unlike disk forensics, network forensics deals with **volatile and dynamic data**, because network packets exist only briefly while in transit.

Network forensics is mainly used for:

- **Security investigations:** Detecting intrusions, attacks, and policy violations.
- **Law enforcement investigations:** Reconstructing emails, file transfers, chat sessions, and identifying suspects.

Because attackers may delete logs on compromised systems, **network-level evidence** can sometimes be the only reliable source of proof.

2. Network Components and Their Forensic Importance

Understanding network components is essential for effective forensic analysis.

2.1 Host

A **host** is any device connected to a network with an IP address, such as a computer or server. Hosts often contain valuable forensic data, including:

- User activity
- Network logs
- Application data

2.2 Node

A **node** is any device participating in network communication. Nodes can be hosts, routers, switches, or other devices. Nodes help investigators understand the **path of data transmission**.

2.3 Router

A **router** forwards packets between networks. Router logs can provide:

- Source and destination IP addresses
- Routing paths
- Evidence for tracing attackers

2.4 Switch

A **switch** connects devices within a local network. It maintains MAC address tables (CAM tables), useful in identifying:

- Which device was connected to which port
- VLAN-related activity

2.5 Hub

A **hub** broadcasts traffic to all connected devices. Though mostly obsolete, hubs allow easier traffic capture because all packets are visible.

2.6 Network Interface Card (NIC)

The **NIC** connects a device to the network and contains the MAC address. Investigators can place the NIC in **promiscuous mode** to capture all passing traffic.

3. OSI and TCP/IP Models

3.1 OSI Model

The **OSI model** consists of seven layers:

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

Each layer plays a role in how data is transmitted and where forensic evidence can be collected.

3.2 TCP/IP Model

The TCP/IP model has **four layers**:

1. Link
2. Network

3. Transport
4. Application

Most real-world network forensics investigations are based on the TCP/IP model.

4. Forensics Information from Networks

Major forensic data sources include:

- Hosts
- Routers
- Firewalls
- Switches
- IDS/IDPS
- Wireless Access Points (WAPs)

Intrusion Detection and Prevention Systems (IDS/IDPS)

IDS and IDPS monitor network traffic and generate logs about:

- Suspicious behavior
- Policy violations
- Known attack patterns

These logs are highly valuable for forensic analysis.

5. Log Analysis

Log analysis is a critical step in network forensics.

5.1 Time Stamp Analysis

- Accurate time is essential for timeline reconstruction.
- Network Time Protocol (NTP) ensures clock synchronization.
- Investigators must verify whether NTP was enabled.

5.2 Data Analysis

Network data is transmitted in packets that may take different paths. TCP sequence numbers and acknowledgments help reconstruct:

- Sessions
- File transfers
- Communication timelines

Protocols commonly analyzed include HTTP, FTP, SMTP, DNS, DHCP, ARP, and SSH.

6. Network Forensics Tools

6.1 Technology-Based Tools

- **Network Taps:** Hardware devices that copy traffic for monitoring.
- **Port Mirroring:** Duplicates switch traffic to a monitoring port.
- **Promiscuous Mode:** Allows NICs to capture all packets.

6.2 Software-Based Tools

Wireshark:

- Packet capture and protocol analysis
- GUI-based filtering and visualization
- Reconstructs sessions and files

Tcpdump:

- Command-line packet analyzer
- Useful for fast, low-level packet inspection
- Often used on servers and routers